# Analysis patterns

## MERSENNES EXPONENTS as SEQUENCES CHAINS LATTICES

**LATTICE BUILDER EXPORT**
==========

Export Date: 3/19/2026, 9:56:02 PM
Total numbers used: 579
Lattices created: 50
Duplicates allowed: 50

**NUMBERS DUPLICATES ALLOWED:**
4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 15, 16, 17, 18, 19, 20, 21, 22, 23, 25, 26, 31, 32, 34, 35, 37, 38, 39, 40, 42, 43, 45, 46, 52, 53, 64, 70, 75, 77, 84, 90, 92, 107, 141, 151, 155, 181, 184, 283, 566

## WE HAVE BASE M1= 2 AND BASE M2= 3

**PATTERN MEANING: 0 = x 2 AND 1 = x 2 +1**
**Numbers n in red color are duplicate(in order of growing sequences)**
**LATTICES:**
---

**M1 = 2** -seed 2
**M2 = 3** -seed 3

### M3 = 5

**Lattice 1** (Pattern: 1)
2 → **5** (×2+1)

### M4 = 7

**Lattice 2** (Pattern: 1)
3 → **7** (×2+1)

### M5 = 13

**Lattice 3** (Pattern: 01)
3 → 6 (×2) → **13** (×2+1)

**B** 3,6, **-13**

3-
6- 0
13-01

### M6 = 17

**Lattice 4** (Pattern: **001**)

2 → 4 (×2) → 8 (×2) → **17** (×2+1)

**B** 2,4,8 -**17**

**2**-
4- 0
8- 00
**17**-001

**M7 = 19**

**Lattice 5** (Pattern: **011**)

Contains duplicates

2 → **4** (×2) → 9 (×2+1) → **19** (×2+1)

**B** 2,4,9 -**19**

**2**-
4- 0
9- 01
**19**-011

**M8 = 31**

**Lattice 6** (Pattern: **111**)

Contains duplicates

**3** → **7** (×2+1) → 15 (×2+1) → **31** (×2+1)

**B** 3,7,15 -**31**

**3**-
**7**- 1
15-11
**31**-111

**M9 = 61**

**Lattice 7** (Pattern: **1101**)

Contains duplicates

**3** → **7** (×2+1) → **15** (×2+1) → 30 (×2) → **61** (×2+1)

**B** 3,7,15,30 -**61**

**3**-
**7**- 1
15-11
30-110
**61**-1101

**M10 = 89**

**Lattice 8** (Pattern: **11001**)
Contains duplicates
2 → **5** (×2+1) → 11 (×2+1) → 22 (×2) → 44 (×2) → **89** (×2+1)

**B** 2,5,11,22,44, -**89**

**2**-
**5**- 1
**11**-11
22-110
44-1100
**89**-11001

## M11 = 107

**Lattice 9** (Pattern: **01011**)
Contains duplicates
**3** → **6** (×2) → **13** (×2+1) → 26 (×2) → 53 (×2+1) → **107** (×2+1)

**B** 3,6,13,26,**53** -**107**

**3**-
6- 0
**13**- 01
26- 010
**53**- 0101
**107**-01011

## M12 = 127

**Lattice 10** (Pattern: **11111**)
Contains duplicates
**3** → **7** (×2+1) → **15** (×2+1) → **31** (×2+1) → 63 (×2+1) → **127** (×2+1)

**B** 3,7,15,31,63 -**127**

**3**-
**7**- 1
15- 11
**31**- 111
63- 1111
**127**-11111

## M13 = 521

**Lattice 11** (Pattern: **00001001**)
Contains duplicates
2 → **4** (×2) → **8** (×2) → 16 (×2) → 32 (×2) → 65 (×2+1) → 130 (×2) → 260 (×2) → **521** (×2+1)

**B** 2,4,8,16,32,65,130,260 -**521**

```
2-
4-   0
8-   00
16-  000
32-  0000
65-  00001
130-000010
260-0000100
521-00001001
```

**M14 = 607**

**Lattice 12** (Pattern: **01011111**)
Contains duplicates
2 → **4** (×2) → **9** (×2+1) → 18 (×2) → 37 (×2+1) → 75 (×2+1) → 151 (×2+1) → 303 (×2+1) → **607** (×2+1)

**B** 2,4,9,18,**37**,75,**151**,303 -**607**

```
2-
4-   0
9-   01
18-  010
37-  0101
75-  01011
151-010111
303-0101111
607-01011111
```

**M15 = 1279**

**Lattice 13** (Pattern: **011111111**)
Contains duplicates
2 → **4** (×2) → **9** (×2+1) → **19** (×2+1) → 39 (×2+1) → 79 (×2+1) → 159 (×2+1) → 319 (×2+1) → 639 (×2+1) → **1279** (×2+1)

**B** 2,4,9,19,39,79,159,319,639 -**1279**

```
2-
4-   0
9-   01
19-  011
39-  0111
79-  01111
159- 011111
319- 0111111
639- 01111111
1279-011111111
```

## M16 = 2203

**Lattice 14** (Pattern: **0010011011**)
Contains duplicates
2 → **4** (×2) → **8** (×2) → **17** (×2+1) → 34 (×2) → 68 (×2) → 137 (×2+1) → 275 (×2+1) → 550 (×2) → 1101 (×2+1) → **2203** (×2+1)

**B** 2,4,8,17,34,68,137,275,550,1101 -**2203**

```
2-
4-   0
8-   00
17-  001
34-  0010
68-  00100
137- 001001
275- 0010011
550- 00100110
1101-001001101
2203-0010011011
```

## M17 = 2281

## Lattice 15 (Pattern: 0011101001)

Contains duplicates

2 → 4 (×2) → 8 (×2) → 17 (×2+1) → 35 (×2+1) → 71 (×2+1) → 142 (×2) → 285 (×2+1) → 570 (×2) → 1140 (×2) → **2281** (×2+1)

**B** 2,4,8,17,35,71,142,285,570,1140 -**2281**

```
2-
4-   0
8-   00
17-  001
35-  0011
71-  00111
142- 001110
285- 0011101
570- 00111010
1140-001110100
2281-0011101001
```

## M18 = 3217

## Lattice 16 (Pattern: 0010010001)

Contains duplicates

3 → 6 (×2) → 12 (×2) → 25 (×2+1) → 50 (×2) → 100 (×2) → 201 (×2+1) → 402 (×2) → 804 (×2) → 1608 (×2) → **3217** (×2+1)

**B** 3,6,12,25,50,100,201,402,804,1608 -**3217**

```
3-
6-   0
12-  00
25-  001
50-  0010
100- 00100
201- 001001
402- 0010010
804- 00100100
1608-001001000
3217-0010010001
```

## M19 = 4253

**Lattice 17** (Pattern: **00010011101**)
Contains duplicates
2 → **4** (×2) → **8** (×2) → **16** (×2) → 33 (×2+1) → 66 (×2) → 132 (×2) → 265 (×2+1) → 531 (×2+1) → 1063 (×2+1) → 2126 (×2) → **4253** (×2+1)

**B** 2,4,8,16,33,66,132,265,531,**1063**,2126 -**4253**

**2**-
4- 0
8- 00
16- 000
33- 0001
66- 00010
132- 000100
265- 0001001
531- 00010011
**1063**-000100111
2126-0001001110
**4253**-00010011101

## M20 = 4423

**Lattice 18** (Pattern: **00101000111**)
Contains duplicates
2 → **4** (×2) → **8** (×2) → **17** (×2+1) → **34** (×2) → 69 (×2+1) → 138 (×2) → 276 (×2) → 552 (×2) → 1105 (×2+1) → 2211 (×2+1) → **4423** (×2+1)

**B** 2,4,8,17,34,69,138,276,552,1105,2211 -**4423**

**2**-
4- 0
8- 00
**17**- 001
34- 0010
69- 00101
138- 001010
276- 0010100
552- 00101000
1105-001010001
2211-0010100011
**4423**-00101000111

## M21 = 9689

## Lattice 19 (Pattern: 010111011001)

Contains duplicates

2 → 4 (×2) → 9 (×2+1) → 18 (×2) → 37 (×2+1) → 75 (×2+1) → 151 (×2+1) → 302 (×2) → 605 (×2+1) → 1211 (×2+1) → 2422 (×2) → 4844 (×2) → **9689** (×2+1)

**B** 2,4,9,18,37,75,151,302,605,1211,2422,4844 -**9689**

```
2-
4-   0
9-   01
18-  010
37-  0101
75-  01011
151- 010111
302- 0101110
605- 01011101
1211-010111011
2422-0101110110
4844-01011101100
9689-010111011001
```

## M22 = 9941

## Lattice 20 (Pattern: 011011010101)

Contains duplicates

2 → 4 (×2) → 9 (×2+1) → 19 (×2+1) → 38 (×2) → 77 (×2+1) → 155 (×2+1) → 310 (×2) → 621 (×2+1) → 1242 (×2) → 2485 (×2+1) → 4970 (×2) → **9941** (×2+1)

**B** 2,4,9,19,38,77,155,310,621,1242,2485,4970 -**9941**

```
2-
4-   0
9-   01
19-  011
38-  0110
77-  01101
155- 011011
310- 0110110
621- 01101101
1242-011011010
2485-0110110101
4970-01101101010
9941-011011010101
```

## M23 = 11213

## Lattice 21 (Pattern: 101111001101)

Contains duplicates

2 → **5** (×2+1) → 10 (×2) → 21 (×2+1) → 43 (×2+1) → 87 (×2+1) → 175 (×2+1) → 350 (×2) → 700 (×2) → 1401 (×2+1) → 2803 (×2+1) → 5606 (×2) → **11213** (×2+1)

**B** 2,5,10,21,**43**,87,175,350,700,1401,**2803**,5606 -**11213**

```
2-
5-     1
10-    10
21-    101
43-    1011
87-    10111
175-   101111
350-   1011110
700-   10111100
1401-  101111001
2803-  1011110011
5606-  10111100110
11213-101111001101
```

## M24 = 19937

## Lattice 22 (Pattern: 0110111100001)

Contains duplicates

2 → **4** (×2) → **9** (×2+1) → **19** (×2+1) → **38** (×2) → **77** (×2+1) → **155** (×2+1) → 311 (×2+1) → 623 (×2+1) → 1246 (×2) → 2492 (×2) → 4984 (×2) → 9968 (×2) → **19937** (×2+1)

**B** 2,4,9,19,38,77,155,**311**,623,1246,2492,4984,9968 -**19937**

```
2-
4-     0
9-     01
19-    011
38-    0110
77-    01101
155-   011011
311-   0110111
623-   01101111
1246-  011011110
2492-  0110111100
4984-  01101111000
9968-  011011110000
19937-0110111100001
```

## M25 = 21701

## Lattice 23 (Pattern: 1010011000101)

Contains duplicates

2 → **5** (×2+1) → **10** (×2) → **21** (×2+1) → 42 (×2) → 84 (×2) → 169 (×2+1) → 339 (×2+1) → 678 (×2) → 1356 (×2) → 2712 (×2) → 5425 (×2+1) → 10850 (×2) → **21701** (×2+1)

**B** 2,5,10,21,42,84,169,339,678,1356,2712,5425,10850 -**21701**

```
2-
5-     1
10-    10
21-    101
42-    1010
84-    10100
169-   101001
339-   1010011
678-   10100110
1356-  101001100
2712-  1010011000
5425-  10100110001
10850-1010011000010
21701-1010011000101
```

## M26 = 23209

## Lattice 24 (Pattern: 1101010101001)

Contains duplicates

2 → **5** (×2+1) → **11** (×2+1) → **22** (×2) → 45 (×2+1) → 90 (×2) → 181 (×2+1) → 362 (×2) → 725 (×2+1) → 1450 (×2) → 2901 (×2+1) → 5802 (×2) → 11604 (×2) → **23209** (×2+1)

**B** 2,5,11,22,45,90,181,362,725,1450,2901,5802,11604 -**23209**

```
2-
5-     1
11-    11
22-    110
45-    1101
90-    11010
181-   110101
362-   1101010
725-   11010101
1450-  110101010
2901-  1101010101
5802-  11010101010
11604-110101010100
23209-1101010101001
```

## M27 = 44497

**Lattice 25** (Pattern: **10110111010001**)
Contains duplicates
2 → **5** (×2+1) → **10** (×2) → **21** (×2+1) → **43** (×2+1) → 86 (×2) → 173 (×2+1) → 347 (×2+1) → 695 (×2+1) → 1390 (×2) → 2781 (×2+1) → 5562 (×2) → 11124 (×2) → 22248 (×2) → **44497** (×2+1)

**B** 2,5,10,21,**43**,86,**173**,**347**,695,1390,2781,5562,11124,22248 -**44497**

```
2-
5-     1
10-    10
21-    101
43-    1011
86-    10110
173-   101101
347-   1011011
695-   10110111
1390-  101101110
2781-  1011011101
5562-  10110111010
11124-101101110100
22248-1011011101000
44497-10110111010001
```

## M28 = 86243

**Lattice 26** (Pattern: **101000011100011**)
Contains duplicates
2 → **5** (×2+1) → **10** (×2) → **21** (×2+1) → **42** (×2) → **84** (×2) → 168 (×2) → 336 (×2) → 673 (×2+1) → 1347 (×2+1) → 2695 (×2+1) → 5390 (×2) → 10780 (×2) → 21560 (×2) → 43121 (×2+1) → **86243** (×2+1)

**B** 2,5,10,21,42,84,168,336,**673**,1347,2695,5390,10780,21560,43121 -**86243**

```
2-
5-     1
10-    10
21-    101
42-    1010
84-    10100
168-   101000
336-   1010000
673-   10100001
1347-  101000011
2695-  1010000111
5390-  10100001110
10780-101000011100
43121-10100001110001
86243-101000011100011
```

## M29 = 110503

## Lattice 27 (Pattern: 010111110100111)

Contains duplicates

3 → 6 (×2) → 13 (×2+1) → 26 (×2) → 53 (×2+1) → 107 (×2+1) → 215 (×2+1) → 431 (×2+1) → 863 (×2+1) → 1726 (×2) → 3453 (×2+1) → 6906 (×2) → 13812 (×2) → 27625 (×2+1) → 55251 (×2+1) → **110503** (×2+1)

**B** 3,6,13,26,53,107,215,431,863,1726,3453,6906,13812,27625,55251 -**110503**

```
3-
6-      0
13-     01
26-     010
53-     0101
107-    01011
215-    010111
431-    0101111
863-    01011111
1726-   010111110
3453-   0101111101
6906-   01011111010
13812-  010111110100
27625-  0101111101001
55251-  01011111010011
110503-010111110100111
```

# M30 = 132049

## Lattice 28 (Pattern: 0000001111010001)

Contains duplicates

2 → 4 (×2) → 8 (×2) → 16 (×2) → 32 (×2) → 64 (×2) → 128 (×2) → 257 (×2+1) → 515 (×2+1) → 1031 (×2+1) → 2063 (×2+1) → 4126 (×2) → 8253 (×2+1) → 16506 (×2) → 33012 (×2) → 66024 (×2) → **132049** (×2+1)

**B** 2,4,8,16,32,64,128,257,515,1031,2063,4126,8253,16506,33012,66024 -**132049**

```
2-
4-      0
8-      00
16-     000
32-     0000
64-     00000
128-    000000
257-    0000001
515-    00000011
1031-   000000111
2063-   0000001111
4126-   00000011110
8253-   000000111101
16506-  0000001111010
33012-  00000011110100
66024-  000000111101000
132049-0000001111010001
```

## M31 = 216091

**Lattice 29** (Pattern: **0100110000011011**)

Contains duplicates

**3** → **6** (×2) → **13** (×2+1) → **26** (×2) → 52 (×2) → 105 (×2+1) → 211 (×2+1) → 422 (×2) → 844 (×2) → 1688 (×2) → 3376 (×2) → 6752 (×2) → 13505 (×2+1) → 27011 (×2+1) → 54022 (×2) → 108045 (×2+1) → **216091** (×2+1)

**B** 3,6,13,26,52,105,211,422,844,1688,3376,6752,13505,**27011**,54022,108045 -**216091**

```
3-
6-      0
13-     01
26-     010
52-     0100
105-    01001
211-    010011
422-    0100110
844-    01001100
1688-   010011000
3376-   0100110000
6752-   01001100000
13505-  010011000001
27011-  0100110000011
54022-  01001100000110
108045-010011000001101
216091-0100110000011011
```

# M32 = 756839

## Lattice 30 (Pattern: 111000110001100111)

Contains duplicates

2 → **5** (×2+1) → **11** (×2+1) → 23 (×2+1) → 46 (×2) → 92 (×2) → 184 (×2) → 369 (×2+1) → 739 (×2+1) → 1478 (×2) → 2956 (×2) → 5912 (×2) → 11825 (×2+1) → 23651 (×2+1) → 47302 (×2) → 94604 (×2) → 189209 (×2+1) → 378419 (×2+1) → **756839** (×2+1)

**B** 2,5,**11**,**23**,46,92,184,369,**739**,1478,2956,5912,11825,23651,47302,94604,189209,378419 -**756839**

**2**-
**5**- 1
**11**- 11
**23**- 111
46- 1110
92- 11100
184- 111000
369- 1110001
**739**- 11100011
1478- 111000110
2956- 1110001100
5912- 11100011000
11825- 111000110001
23651- 1110001100011
47302- 11100011000110
94604- 111000110001100
189209-1110001100011001
378419-11100011000110011
**756839**-111000110001100111

## M33 = 859433

**Lattice 31** (Pattern: **010001110100101001**)

Contains duplicates

**3** → **6** (×2) → **13** (×2+1) → **26** (×2) → **52** (×2) → 104 (×2) → 209 (×2+1) → 419 (×2+1) → 839 (×2+1) → 1678 (×2) → 3357 (×2+1) → 6714 (×2) → 13428 (×2) → 26857 (×2+1) → 53714 (×2) → 107429 (×2+1) → 214858 (×2) → 429716 (×2) → **859433** (×2+1)

**B** 3,6,13,26,52,104,209,**419**,**839**,1678,3357,6714,13428,26857,53714,107429, 214858, 429716 -**859433**

**3**-
6- 0
**13**- 01
26- 010
52- 0100
104- 01000
209- 010001
**419**- 0100011
**839**- 01000111
1678- 010001110
3357- 0100011101
6714- 01000111010
13428- 010001110100
26857- 0100011101001
53714- 01000111010010
107429-010001110100101
214858-0100011101001010
429716-01000111010010100
**859433**-010001110100101001

# M34 = 1257787

## Lattice 32 (Pattern: 0110011000100111011)

Contains duplicates

2 → **4** (×2) → **9** (×2+1) → **19** (×2+1) → **38** (×2) → 76 (×2) → 153 (×2+1) → 307 (×2+1) → 614 (×2) → 1228 (×2) → 2456 (×2) → 4913 (×2+1) → 9826 (×2) → 19652 (×2) → 39305 (×2+1) → 78611 (×2+1) → 157223 (×2+1) → 314446 (×2) → 628893 (×2+1) → **1257787** (×2+1)

**B** 2,4,9,19,38,76,153,**307**,614,1228,2456,4913,9826,19652,39305,78611,157223, 314446, 628893 -**1257787**

| | |
|---|---|
| **2**- | |
| 4- | 0 |
| 9- | 01 |
| **19**- | 011 |
| 38- | 0110 |
| 76- | 01100 |
| 153- | 011001 |
| **307**- | 0110011 |
| 614- | 01100110 |
| 1228- | 011001100 |
| 2456- | 0110011000 |
| 4913- | 01100110001 |
| 9826- | 011001100010 |
| 19652- | 0110011000100 |
| 39305- | 01100110001001 |
| 78611- | 011001100010011 |
| 157223- | 0110011000100111 |
| 314446- | 01100110001001110 |
| 628893- | 011001100010011101 |
| **1257787**- | 0110011000100111011 |

# M35 = 1398269

## Lattice 33 (Pattern: 1010101010111111101)

Contains duplicates

2 → **5** (×2+1) → **10** (×2) → **21** (×2+1) → **42** (×2) → 85 (×2+1) → 170 (×2) → 341 (×2+1) → 682 (×2) → 1365 (×2+1) → 2730 (×2) → 5461 (×2+1) → 10923 (×2+1) → 21847 (×2+1) → 43695 (×2+1) → 87391 (×2+1) → 174783 (×2+1) → 349567 (×2+1) → 699134 (×2) → **1398269** (×2+1)

**B** 2,5,10,21,42,85,170,341,682,1365,2730,5461,10923,21847,43695,87391, 174783, **349567**,699134 -**1398269**

| | |
|---|---|
| **2**- | |
| **5**- | 1 |
| 10- | 10 |
| 21- | 101 |
| 42- | 1010 |
| 85- | 10101 |
| 170- | 101010 |
| 341- | 1010101 |
| 682- | 10101010 |
| 1365- | 101010101 |
| 2730- | 1010101010 |
| 5461- | 10101010101 |
| 10923- | 101010101011 |
| 21847- | 1010101010111 |
| 43695- | 10101010101111 |
| 87391- | 101010101011111 |
| 174783- | 1010101010111111 |
| **349567**- | 10101010101111111 |
| 699134- | 101010101011111110 |
| **1398269**- | 1010101010111111101 |

# M36 = 2976221

## Lattice 34 (Pattern: 11010110100111011101)

Contains duplicates

2 → **5** (×2+1) → **11** (×2+1) → **22** (×2) → **45** (×2+1) → **90** (×2) → **181** (×2+1) → 363 (×2+1) → 726 (×2) → 1453 (×2+1) → 2906 (×2) → 5812 (×2) → 11625 (×2+1) → 23251 (×2+1) → 46503 (×2+1) → 93006 (×2) → 186013 (×2+1) → 372027 (×2+1) → 744055 (×2+1) → 1488110 (×2) → **2976221** (×2+1)

**B** 2,5,**11**,22,45,90,**181**,363,726,**1453**,2906,5812,11625,**23251**,46503,93006,**186013**, 372027,744055,1488110 – **2976221**

| | |
|---|---|
| **2**- | |
| **5**- | 1 |
| **11**- | 11 |
| 22- | 110 |
| 45- | 1101 |
| 90- | 11010 |
| **181**- | 110101 |
| 363- | 1101011 |
| 726- | 11010110 |
| **1453**- | 110101101 |
| 2906- | 1101011010 |
| 5812- | 11010110100 |
| 11625- | 110101101001 |
| **23251-** | 1101011010011 |
| 46503- | 11010110100111 |
| 93006- | 110101101001110 |
| **186013**- | 1101011010011101 |
| 372027- | 11010110100111011 |
| 744055- | 110101101001110111 |
| 1488110– | 1101011010011101110 |
| **2976221**- | 11010110100111011101 |

# M37 = 3021377

## Lattice 35 (Pattern: 11100001101001000001)

Contains duplicates

2 → **5** (×2+1) → **11** (×2+1) → **23** (×2+1) → **46** (×2) → **92** (×2) → **184** (×2) → 368 (×2) → 737 (×2+1) → 1475 (×2+1) → 2950 (×2) → 5901 (×2+1) → 11802 (×2) → 23604 (×2) → 47209 (×2+1) → 94418 (×2) → 188836 (×2) → 377672 (×2) → 755344 (×2) → 1510688 (×2) → **3021377** (×2+1)

**B** 2,5,11,23,46,92,184,368,737,1475,2950,5901,11802,23604,47209,94418,188836,377672,755344,1510688 -**3021377**

| | |
|---|---|
| **2**- | |
| **5**- | 1 |
| **11**- | 11 |
| **23**- | 111 |
| 46- | 1110 |
| 92- | 11100 |
| 184- | 111000 |
| 368- | 1110000 |
| 737- | 11100001 |
| 1475- | 111000011 |
| 2950- | 1110000110 |
| 5901- | 11100001101 |
| 11802- | 111000011010 |
| 23604- | 1110000110100 |
| 47209- | 11100001101001 |
| 94418- | 111000011010010 |
| 188836- | 1110000110100100 |
| 377672- | 11100001101001000 |
| 755344- | 111000011010010000 |
| 1510688- | 1110000110100100000 |
| **3021377**- | 11100001101001000001 |

## M38 = 6972593

**Lattice 36** (Pattern: **01010011001001011000 1**)

Contains duplicates

**3** → **6** (×2) → **13** (×2+1) → **26** (×2) → **53** (×2+1) → 106 (×2) → 212 (×2) → 425 (×2+1) → 851 (×2+1) → 1702 (×2) → 3404 (×2) → 6809 (×2+1) → 13618 (×2) → 27236 (×2) → 54473 (×2+1) → 108946 (×2) → 217893 (×2+1) → 435787 (×2+1) → 871574 (×2) → 1743148 (×2) → 3486296 (×2) → **6972593** (×2+1)

**B** 3,6,13,26,53,106,212,425,851,1702,3404,6809,13618,27236,54473,108946, 217893, 435787,871574,1743148,3486296 -**6972593**

| | |
|---|---|
| **3**- | |
| 6- | 0 |
| **13**- | 01 |
| 26- | 010 |
| **53**- | 0101 |
| 106- | 01010 |
| 212- | 010100 |
| 425- | 0101001 |
| 851- | 01010011 |
| 1702- | 010100110 |
| 3404- | 0101001100 |
| 6809- | 01010011001 |
| 13618- | 010100110010 |
| 27236- | 0101001100100 |
| 54473- | 01010011001001 |
| 108946- | 010100110010010 |
| 217893- | 0101001100100101 |
| 435787- | 01010011001001011 |
| 871574- | 010100110010010110 |
| 1743148- | 0101001100100101100 |
| 3486296- | 01010011001001011000 |
| **6972593**- | 010100110010010110001 |

## M39 = 13466917

**Lattice 37** (Pattern: **00110101111101001001 01**)

Contains duplicates

**3** → **6** (×2) → **12** (×2) → **25** (×2+1) → 51 (×2+1) → 102 (×2) → 205 (×2+1) → 410 (×2) → 821 (×2+1) → 1643 (×2+1) → 3287 (×2+1) → 6575 (×2+1) → 13151 (×2+1) → 26302 (×2) → 52605 (×2+1) → 105210 (×2) → 210420 (×2) → 420841 (×2+1) → 841682 (×2) → 1683364 (×2) → 3366729 (×2+1) → 6733458 (×2) → **13466917** (×2+1)

**B** 3,6,12,25,51,102,205,410,**821**,1643,3287,6575,**13151**,26302,52605,105210, 210420, 420841,841682,1683364,3366729,6733458 -**13466917**

| | |
|---|---|
| **3**- | |
| 6- | 0 |
| 12- | 00 |
| 25- | 001 |
| 51- | 0011 |
| 102- | 00110 |
| 205- | 001101 |
| 410- | 0011010 |
| **821**- | 00110101 |
| 1643- | 001101011 |
| 3287- | 0011010111 |
| 6575- | 00110101111 |
| **13151**- | 001101011111 |
| 26302- | 0011010111110 |
| 52605- | 00110101111101 |
| 105210- | 001101011111010 |
| 210420- | 0011010111110100 |
| 420841- | 00110101111101001 |
| 841682- | 001101011111010010 |
| 1683364- | 0011010111110100100 |
| 3366729- | 00110101111101001001 |
| 6733458- | 001101011111010010010 |
| **13466917**- | 0011010111110100100101 |

# M40 = 20996011

## Lattice 38 (Pattern: 1000000010111111010101 1)

Contains duplicates

2 → **5** (×2+1) → **10** (×2) → 20 (×2) → 40 (×2) → 80 (×2) → 160 (×2) → 320 (×2) → 640 (×2) → 1281 (×2+1) → 2562 (×2) → 5125 (×2+1) → 10251 (×2+1) → 20503 (×2+1) → 41007 (×2+1) → 82015 (×2+1) → 164031 (×2+1) → 328062 (×2) → 656125 (×2+1) → 1312250 (×2) → 2624501 (×2+1) → 5249002 (×2) → 10498005 (×2+1) → **20996011** (×2+1)

**B** 2,5,10,20,40,80,160,320,640,1281,2562,5125,10251,20503,41007,82015,164031,328062,656125,1312250,2624501,5249002,10498005 -**20996011**

| | |
|---|---|
| **2**- | |
| **5**- | 1 |
| 10- | 10 |
| 20- | 100 |
| 40- | 1000 |
| 80- | 10000 |
| 160- | 100000 |
| 320- | 1000000 |
| 640- | 10000000 |
| 1281- | 100000001 |
| 2562- | 1000000010 |
| 5125- | 10000000101 |
| 10251- | 100000001011 |
| 20503- | 1000000010111 |
| 41007- | 10000000101111 |
| 82015- | 100000001011111 |
| 164031- | 1000000010111111 |
| 328062- | 10000000101111110 |
| 656125- | 100000001011111101 |
| 1312250- | 1000000010111111010 |
| 2624501- | 10000000101111110101 |
| 5249002- | 100000001011111101010 |
| 10498005- | 1000000010111111010101 |
| **20996011**- | 10000000101111110101011 |

# M41 = 24036583

## Lattice 39 (Pattern: 1101110110001001110011 1)

Contains duplicates

2 → **5** (×2+1) → **11** (×2+1) → **22** (×2) → **45** (×2+1) → 91 (×2+1) → 183 (×2+1) → 366 (×2) → 733 (×2+1) → 1467 (×2+1) → 2934 (×2) → 5868 (×2) → 11736 (×2) → 23473 (×2+1) → 46946 (×2) → 93892 (×2) → 187785 (×2+1) → 375571 (×2+1) → 751143 (×2+1) → 1502286 (×2) → 3004572 (×2) → 6009145 (×2+1) → 12018291 (×2+1) → **24036583** (×2+1)

**B** 2,5,**11**,22,45,91,183,366,**733**,1467,2934,5868,11736,**23473**, 46946,93892,187785, 375571,751143, 1502286,3004572,6009145,12018291 -**24036583**

| | |
|---|---|
| **2**- | |
| **5**- | 1 |
| **11**- | 11 |
| 22- | 110 |
| 45- | 1101 |
| 91- | 11011 |
| 183- | 110111 |
| 366- | 1101110 |
| **733**- | 11011101 |
| 1467- | 110111011 |
| 2934- | 1101110110 |
| 5868- | 11011101100 |
| 11736- | 110111011000 |
| **23473**- | 1101110110001 |
| 46946- | 11011101100010 |
| 93892- | 110111011000100 |
| 187785- | 1101110110001001 |
| 375571- | 11011101100010011 |
| 751143- | 110111011000100111 |
| 1502286- | 1101110110001001110 |
| 3004572- | 11011101100010011100 |
| 6009145- | 110111011000100111001 |
| 12018291- | 1101110110001001110011 |
| **24036583**- | 11011101100010011100111 |

## M42 = 25964951

**Lattice 40** (Pattern: **000110000110001100101 11**)

Contains duplicates

**3** → **6** (×2) → **12** (×2) → 24 (×2) → 49 (×2+1) → 99 (×2+1) → 198 (×2) → 396 (×2) → 792 (×2) → 1584 (×2) → 3169 (×2+1) → 6339 (×2+1) → 12678 (×2) → 25356 (×2) → 50712 (×2) → 101425 (×2+1) → 202851 (×2+1) → 405702 (×2) → 811404 (×2) → 1622809 (×2+1) → 3245618 (×2) → 6491237 (×2+1) → 12982475 (×2+1) → **25964951** (×2+1)

**B** 3,6,12,24,49,99,198,396,792,1584,**3169**,6339,12678,25356,50712,101425,202851, 405702,811404,1622809,3245618, 6491237,12982475 -**25964951**

| | |
|---|---|
| **3**- | |
| 6- | 0 |
| 12- | 00 |
| 24- | 000 |
| 49- | 0001 |
| 99- | 00011 |
| 198- | 000110 |
| 396- | 0001100 |
| 792- | 00011000 |
| 1584- | 000110000 |
| **3169**- | 0001100001 |
| 6339- | 00011000011 |
| 12678- | 000110000110 |
| 25356- | 0001100001100 |
| 50712- | 00011000011000 |
| 101425- | 000110000110001 |
| 202851- | 0001100001100011 |
| 405702- | 00011000011000110 |
| 811404- | 000110000110001100 |
| 1622809- | 0001100001100011001 |
| 3245618- | 00011000011000110010 |
| 6491237- | 000110000110001100101 |
| 12982475- | 0001100001100011001011 |
| **25964951**- | 0001100001100011001011 1 |

## M43 = 30402457

**Lattice 41** (Pattern: **1001111111001111001100 1**)

Contains duplicates

**3** → **7** (×2+1) → 14 (×2) → 28 (×2) → 57 (×2+1) → 115 (×2+1) → 231 (×2+1) → 463 (×2+1) → 927 (×2+1) → 1855 (×2+1) → 3711 (×2+1) → 7422 (×2) → 14844 (×2) → 29689 (×2+1) → 59379 (×2+1) → 118759 (×2+1) → 237519 (×2+1) → 475038 (×2) → 950076 (×2) → 1900153 (×2+1) → 3800307 (×2+1) → 7600614 (×2) → 15201228 (×2) → **30402457** (×2+1)

**B** 3,7,14,28,57,115,231,**463**,927,1855,3711,7422,14844,29689,59379,118759, 237519, 475038,950076,**1900153**,3800307,7600614,15201228 -**30402457**

| | |
|---|---|
| **3**- | |
| **7**- | 1 |
| 14- | 10 |
| 28- | 100 |
| 57- | 1001 |
| 115- | 10011 |
| 231- | 100111 |
| **463**- | 1001111 |
| 927- | 10011111 |
| 1855- | 100111111 |
| 3711- | 1001111111 |
| 7422- | 10011111110 |
| 14844- | 100111111100 |
| 29689- | 1001111111001 |
| 59379- | 10011111110011 |
| 118759- | 100111111100111 |
| 237519- | 1001111111001111 |
| 475038- | 10011111110011110 |
| 950076- | 100111111100111100 |
| **1900153**- | 1001111111001111001 |
| 3800307- | 10011111110011110011 |
| 7600614- | 100111111100111100110 |
| 15201228- | 1001111111001111001100 |
| **30402457**- | 10011111110011110011001 |

## M44 = 32582657

**Lattice 42** (Pattern: **11100010010110000000001**)

Contains duplicates

**3** → **7** (×2+1) → **15** (×2+1) → **31** (×2+1) → 62 (×2) → 124 (×2) → 248 (×2) → 497 (×2+1) → 994 (×2) → 1988 (×2) → 3977 (×2+1) → 7954 (×2) → 15909 (×2+1) → 31819 (×2+1) → 63638 (×2) → 127276 (×2) → 254552 (×2) → 509104 (×2) → 1018208 (×2) → 2036416 (×2) → 4072832 (×2) → 8145664 (×2) → 16291328 (×2) → **32582657** (×2+1)

**B** 3,7,15,31,62,124,248,497,994,1988,3977,7954, 15909,31819,63638,127276,254552, 509104,1018208,2036416,4072832,8145664, 16291328 - **32582657**

| | |
|---|---|
| **3**- | |
| **7**- | 1 |
| 15- | 11 |
| **31**- | 111 |
| 62- | 1110 |
| 124- | 11100 |
| 248- | 111000 |
| 497- | 1110001 |
| 994- | 11100010 |
| 1988- | 111000100 |
| 3977- | 1110001001 |
| 7954- | 11100010010 |
| 15909- | 111000100101 |
| 31819- | 1110001001011 |
| 63638- | 11100010010110 |
| 127276- | 111000100101100 |
| 254552- | 1110001001011000 |
| 509104- | 11100010010110000 |
| 1018208- | 111000100101100000 |
| 2036416- | 1110001001011000000 |
| 4072832- | 11100010010110000000 |
| 8145664- | 111000100101100000000 |
| 16291328- | 1110001001011000000000 |
| **32582657**- | 11100010010110000000001 |

# M45 = 37156667

## Lattice 43 (Pattern: 001101101111011100111011)

Contains duplicates

2 → **4** (×2) → **8** (×2) → **17** (×2+1) → **35** (×2+1) → 70 (×2) → 141 (×2+1) → 283 (×2+1) → 566 (×2) → 1133 (×2+1) → 2267 (×2+1) → 4535 (×2+1) → 9071 (×2+1) → 18142 (×2) → 36285 (×2+1) → 72571 (×2+1) → 145143 (×2+1) → 290286 (×2) → 580572 (×2) → 1161145 (×2+1) → 2322291 (×2+1) → 4644583 (×2+1) → 9289166 (×2) → 18578333 (×2+1) → **37156667** (×2+1)

**B** 2,4,8,17,35,70,141,**283**,566,1133,**2267**,4535,9071,18142,36285,72571,145143, 290286, 580572,1161145,2322291,**4644583**,9289166,18578333 -**37156667**

```
2-
4-        0
8-        00
17-       001
35-       0011
70-       00110
141-      001101
283-      0011011
566-      00110110
1133-     001101101
2267-     0011011011
4535-     00110110111
9071-     001101101111
18142-    0011011011110
36285-    00110110111101
72571-    001101101111011
145143-   0011011011110111
290286-   00110110111101110
580572-   001101101111011100
1161145-  0011011011110111001
2322291-  00110110111101110011
4644583-  001101101111011100111
9289166-  0011011011110111001110
18578333-00110110111101110011101
37156667-001101101111011100111011
```

# M46 = 42643801

## Lattice 44 (Pattern: 10001010101100010101011001)

Contains duplicates

2 → **5** (×2+1) → **10** (×2) → **20** (×2) → **40** (×2) → 81 (×2+1) → 162 (×2) → 325 (×2+1) → 650 (×2) → 1301 (×2+1) → 2602 (×2) → 5205 (×2+1) → 10411 (×2+1) → 20822 (×2) → 41644 (×2) → 83288 (×2) → 166577 (×2+1) → 333154 (×2) → 666309 (×2+1) → 1332618 (×2) → 2665237 (×2+1) → 5330475 (×2+1) → 10660950 (×2) → 21321900 (×2) → **42643801** (×2+1)

**B** 2,5,10,20,40,81,162,325,650,**1301**,2602,5205,10411,20822,41644,83288,166577,333154,666309,1332618,**2665237**,5330475,10660950,21321900 -**42643801**

```
2-
5-        1
10-       10
20-       100
40-       1000
81-       10001
162-      100010
325-      1000101
650-      10001010
1301-     100010101
2602-     1000101010
5205-     10001010101
10411-    100010101011
20822-    1000101010110
41644-    10001010101100
83288-    100010101011000
166577-   1000101010110001
333154-   10001010101100010
666309-   100010101011000101
1332618-  1000101010110001010
2665237-  10001010101100010101
5330475-  100010101011000101011
10660950-1000101010110001010110
21321900-10001010101100010101100
42643801-100010101011000101011001
```

# M47 = 43112609

## Lattice 45 (Pattern: 1001000111011000101000001)

Contains duplicates

2 → **5** (×2+1) → **10** (×2) → **20** (×2) → 41 (×2+1) → 82 (×2) → 164 (×2) → 328 (×2) → 657 (×2+1) → 1315 (×2+1) → 2631 (×2+1) → 5262 (×2) → 10525 (×2+1) → 21051 (×2+1) → 42102 (×2) → 84204 (×2) → 168408 (×2) → 336817 (×2+1) → 673634 (×2) → 1347269 (×2+1) → 2694538 (×2) → 5389076 (×2) → 10778152 (×2) → 21556304 (×2) → **43112609** (×2+1)

**B** 2,5,10,20,**41**,82,164,328,657,1315,2631,5262,10525,21051,42102,84204,168408,336817,673634,1347269,2694538,5389076,10778152,21556304 -**43112609**

| | |
|---|---|
| **2**- | |
| **5**- | 1 |
| 10- | 10 |
| 20- | 100 |
| **41**- | 1001 |
| 82- | 10010 |
| 164- | 100100 |
| 328- | 1001001 |
| 657- | 10100011 |
| 1315- | 100100011 |
| 2631- | 1001000111 |
| 5262- | 10010001110 |
| 10525- | 100100011101 |
| 21051- | 1001000111011 |
| 42102- | 10010001110110 |
| 84204- | 100100011101100 |
| 168408- | 1001000111011000 |
| 336817- | 10010001110110001 |
| 673634- | 100100011101100010 |
| 1347269- | 1001000111011000101 |
| 2694538- | 10010001110110001010 |
| 5389076- | 100100011101100010100 |
| 10778152- | 1001000111011000101000 |
| 21556304- | 10010001110110001010000 |
| **43112609**- | 100100011101100010100001 |

## M48 = 57885161

**Lattice 46** (Pattern: **011100110100000111101001**)

Contains duplicates

**3** → **6** (×2) → **13** (×2+1) → 27 (×2+1) → 55 (×2+1) → 110 (×2) → 220 (×2) → 441 (×2+1) → 883 (×2+1) → 1766 (×2) → 3533 (×2+1) → 7066 (×2) → 14132 (×2) → 28264 (×2) → 56528 (×2) → 113056 (×2) → 226113 (×2+1) → 452227 (×2+1) → 904455 (×2+1) → 1808911 (×2+1) → 3617822 (×2) → 7235645 (×2+1) → 14471290 (×2) → 28942580 (×2) → **57885161** (×2+1)

**B** 3,6,13,27,55,110,220,441,**883**,1766,**3533**,7066,14132,28264, 56528,113056,226113, **452227**,904455, 1808911, 3617822,7235645,14471290,28942580 -**57885161**

| | |
|---|---|
| **3**- | |
| 6- | 0 |
| **13**- | 01 |
| 27- | 011 |
| 55- | 0111 |
| 110- | 01110 |
| 220- | 011100 |
| 441- | 0111001 |
| **883**- | 01110011 |
| 1766- | 011100110 |
| **3533**- | 0111001101 |
| 7066- | 01110011010 |
| 14132- | 011100110100 |
| 28264- | 0111001101000 |
| 56528- | 01110011010000 |
| 113056- | 011100110100000 |
| 226113- | 0111001101000001 |
| **452227**- | 01110011010000011 |
| 904455- | 011100110100000111 |
| 1808911- | 0111001101000001111 |
| 3617822- | 01110011010000011110 |
| 7235645- | 011100110100000111101 |
| 14471290- | 0111001101000001111010 |
| 28942580- | 01110011010000011110100 |
| **57885161**- | 011100110100000111101001 |

# M49 = 74207281

## Lattice 47 (Pattern: 00110110001010000000110001)

Contains duplicates

2 → **4** (×2) → **8** (×2) → **17** (×2+1) → **35** (×2+1) → **70** (×2) → **141** (×2+1) → **283** (×2+1) → **566** (×2) → 1132 (×2) → 2264 (×2) → 4529 (×2+1) → 9058 (×2) → 18117 (×2+1) → 36234 (×2) → 72468 (×2) → 144936 (×2) → 289872 (×2) → 579744 (×2) → 1159488 (×2) → 2318977 (×2+1) → 4637955 (×2+1) → 9275910 (×2) → 18551820 (×2) → 37103640 (×2) → **74207281** (×2+1)

89872,579744,1159488,2318977,4637955,9275910, 18551820, 37103640 are C-)

**B** 2,4,8,17,35,70,141,**283**,566,1132,2264,4529,9058,18117,36234,72468,144936, 289872, 579744,1159488,2318977,4637955,9275910, 18551820, 37103640 -**74207281**

| | |
|---|---|
| **2**- | 1 |
| 4- | 0 |
| 8- | 00 |
| **17**- | 001 |
| 35- | 0011 |
| 70- | 00110 |
| 141- | 001101 |
| **283**- | 0011011 |
| 566- | 00110110 |
| 1132- | 001101100 |
| 2264- | 0011011000 |
| 4529- | 00110110001 |
| 9058- | 001101100010 |
| 18117- | 0011011000101 |
| 36234- | 00110110001010 |
| 72468- | 001101100010100 |
| 144936- | 0011011000101000 |
| 289872- | 00110110001010000 |
| 579744- | 001101100010100000 |
| 1159488- | 0011011000101000000 |
| 2318977- | 00110110001010000001 |
| 4637955- | 001101100010100000011 |
| 9275910- | 0011011000101000000110 |
| 18551820- | 00110110001010000001100 |
| 37103640- | 001101100010100000011000 |
| **74207281**- | 0011011000101000000110001 |

# M50 = 77232917

## Lattice 48 (Pattern: 0100110100111101100010101)

Contains duplicates

2 → **4** (×2) → **9** (×2+1) → **18** (×2) → 36 (×2) → 73 (×2+1) → 147 (×2+1) → 294 (×2) → 589 (×2+1) → 1178 (×2) → 2356 (×2) → 4713 (×2+1) → 9427 (×2+1) → 18855 (×2+1) → 37711 (×2+1) → 75422 (×2) → 150845 (×2+1) → 301691 (×2+1) → 603382 (×2) → 1206764 (×2) → 2413528 (×2) → 4827057 (×2+1) → 9654114 (×2) → 19308229 (×2+1) → 38616458 (×2) → **77232917** (×2+1)

**B** 2,4,9,18,36,**73**,147,294,589,1178,2356,4713,9427,18855,37711,75422,150845,301691,603382,1206764,2413528,4827057,9654114,19308229,3861458 -**77232917**

**2**-
4- 0
9- 01
18- 010
36- 0100
**73**- 01001
147- 101110
294- 0100110
589- 01001101
1178- 010011010
2356- 0100110100
4713- 01001101001
9427- 010011010011
18855- 0100110100111
37711- 01001101001111
75422- 010011010011110
150845- 0100110100111101
301691, 01001101001111011
603382- 010011010011110110
1206764- 0100110100111101100
2413528- 01001101001111011000
4827057- 010011010011110110001
9654114- 0100110100111101100010
19308229-01001101001111011000101
38616458-010011010011110110001010
**77232917**-0100110100111101100010101

# M51 = 82589933

## Lattice 49 (Pattern: 011101100001110001110110 1)

Contains duplicates

2 → **4** (×2) → **9** (×2+1) → **19** (×2+1) → **39** (×2+1) → 78 (×2) → 157 (×2+1) → 315 (×2+1) → 630 (×2) → 1260 (×2) → 2520 (×2) → 5040 (×2) → 10081 (×2+1) → 20163 (×2+1) → 40327 (×2+1) → 80654 (×2) → 161308 (×2) → 322616 (×2) → 645233 (×2+1) → 1290467 (×2+1) → 2580935 (×2+1) → 5161870 (×2) → 10323741 (×2+1) → 20647483 (×2+1) → 41294966 (×2) → **82589933** (×2+1)

**B** 2,4,9,19,39,78,**157**,315,630,1260,2520,5040,10081,20163,40327,80654,161308,322616,**645233**,**1290467**,2580935,5161870,10323741,20647483,41294966 -**82589933**

| | |
|---|---|
| **2**- | |
| 4- | 0 |
| 9- | 01 |
| **19**- | 011 |
| 39- | 0111 |
| 78- | 01110 |
| **157**- | 011101 |
| 315- | 0111011 |
| 630- | 01110110 |
| 1260- | 011101100 |
| 2520- | 0111011000 |
| 5040- | 01110110000 |
| 10081- | 011101100001 |
| 20163- | 0111011000011 |
| 40327- | 01110110000111 |
| 80654- | 011101100001110 |
| 161308- | 0111011000011100 |
| 322616- | 01110110000111000 |
| **645233**- | 011101100001110001 |
| **1290467**- | 0111011000011100011 |
| 2580935- | 01110110000111000111 |
| 5161870- | 011101100001110001110 |
| 10323741- | 0111011000011100011101 |
| 20647483- | 01110110000111000111011 |
| 41294966- | 011101100001110001110110 |
| **82589933**- | 0111011000011100011101101 |

# M52 = 136279841

**Lattice 50** (Pattern: **000001111101111011100100001**)

Contains duplicates

2 → **4** (×2) → **8** (×2) → **16** (×2) → **32** (×2) → **64** (×2) → 129 (×2+1) → 259 (×2+1) → 519 (×2+1) → 1039 (×2+1) → 2079 (×2+1) → 4158 (×2) → 8317 (×2+1) → 16635 (×2+1) → 33271 (×2+1) → 66542 (×2) → 133085 (×2+1) → 266171 (×2+1) → 532343 (×2+1) → 1064686 (×2) → 2129372 (×2) → 4258745 (×2+1) → 8517490 (×2) → 17034980 (×2) → 34069960 (×2) → 68139920 (×2) → **136279841** (×2+1)

**B** 2,4,8,16,32,64,129,259,519,**1039**,2079,4158,**8317**,16635,33271,66542,133085, 266171, 532343,1064686, 2129372, 4258745, 8517490, 17034980, 34069960, 68139920 – **136279841**

| | |
|---|---|
| **2**- | |
| 4- | 0 |
| 8- | 00 |
| 16- | 000 |
| 32- | 0000 |
| 64- | 00000 |
| 129- | 000001 |
| 259- | 0000011 |
| 519- | 00000111 |
| **1039**- | 000001111 |
| 2079- | 0000011111 |
| 4158- | 00000111110 |
| **8317**- | 000001111101 |
| 16635- | 0000011111011 |
| 33271- | 00000111110111 |
| 66542- | 000001111101110 |
| 133085- | 0000011111011101 |
| 266171- | 00000111110111011 |
| 532343- | 000001111101110111 |
| 1064686- | 0000011111011101110 |
| 2129372- | 00000111110111011100 |
| 4258745- | 000001111101110111001 |
| 8517490- | 0000011111011101110010 |
| 17034980- | 00000111110111011100100 |
| 34069960- | 000001111101110111001000 |
| 68139920– | 0000011111011101110010000 |
| **136279841**- | 00000111110111011100100001 |